





CS432
Compiler Design

University of Tripoli
Faculty of Science
Computer Science Department
Fall 2026/2027
by
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1

Agenda

01	Course Overview
02	Course Syllabus
03	Course Evaluation criteria
04	Schedule and Important dates
05	Final Considerations

2

Course Overview

- Course Description:

  Compiler Design will teach students the fundamental concepts and techniques used for building a simple compiler. Focusing on both theory and practice, we will use a sample language to explore the lexical, syntactic and semantic structures of programming languages, and learn to use those structures in implementing a demonstrative compiler. The discussion will also include the examination of intermediate code states, machine code optimisation techniques and support for advanced language features.

3

Course Overview

- Course Outcome:

  At the end of the course, students will understand different considerations and phases of compilation, the impact of language attributes upon the compilation process, the effect of hardware feature on the generated code and the practical fundamentals of compiler implementation.

- Prerequisites:
  - CS241/CS441 Automata Theory and Formal Languages
  - CS336 Concepts of Programming Languages

4

Course Syllabus

- The course content is distributed over 14 weeks as follows:
  - Week 1- Introduction to Compilers: Overview of Compilation, Analysis of source program, phases of compiler.
  - Week 2,3 - Lexical Analysis: Specification and recognizing of tokens, Regular Expressions, Implementation of Lexical Analyzer from DFA. Using The Lexical-Analyzer Generator (Lex).
  - Week 4,5 - Syntax Analysis: Overview of Parsing, Types of parsing including Top-down parsing, Bottom-up parsing, Operator-precedence parsing, LR parsers, The Parser Generator (Yacc).
  - Week 6,7 – Syntax Direct Translation: Syntax-Directed definition, Dependency Graphs, Applications of Syntax-Directed Translation, Syntax-Directed Translation Schemes, Implementing L-Attributed SDD's and generation of symbol table.

5

Course Syllabus

- Continue:
  - Week 8,9 - Intermediate-Code Generation: Three-address code for expressions and statements including assignments, conditionals, loops, procedure calls and generation of temporary variables and tables.
  - Week 10,11 - Code Optimization: Principal sources of optimization, Introduction to data flow analysis and equations, code improving transformations.
  - Week 12,14 – Course Project.

6

Course Evaluation

- Midterm exam :
  Written exam with 5 comprehension questions covering Chapters 1 to 5.
  Worth 20% of Total Grade.
- Compiler project (worth 60% of Total Grade):
  Development of a compiler for a simple language. The project will be evaluated on the following criteria:
  - 20% For correctly compiling and reporting errors of 5 programs.
  - 20% Discussion and prove of comprehension of the code and the application of the process (individual oral exam)
  - 20% Presentation, documentation and punctuality
- Final exam:
  Written exam containing 8 comprehension questions covering the entire course content. Worth 20% of Total Grade.

7

Course Evaluation

- Course Project:
  - 2 to 3 participants in each project team.
  - On Week 2 a complete description of the language to be developed will be handed to each team.
  - On Week 7 (Project Phase 1) a presentation of the characteristics of the language indicating subsets of the language included in project scope, language Grammer (BNF) and Parser code generating a parse tree for language subset chosen.
  - On Week 14 (Project Phase 2) handover for the compiler program source code, complete program documentation and presentation of the project language.
  - On Week 15 Individual project discussion and Test using provided code.

Documents and Templates will be provided during the semester.

8

Schedule and important dates

- Weekly schedule
  - Sunday, from 12:00 to 14:00, Auditorium 70.
  - Tuesday. from 8:00 to 10:00, Auditorium 70.
- Important dates:
  - 04/10/2026: Handout of Project documents
  - 15/11/2026, Before 23:59, MS Teams: Delivery of Phase 1
  - 15/11/2026, 12:00-14:00, Auditorium 70 : Midterm Exam
  - 27/12/2026, Before 23:59, MS Teams: Delivery of Phase 2(Final)
  - 29/12/2026: Project discussion.
  - 04/01/2027, 9:00-11:00: Final Exam

9

Final Considerations

- Strong background in the following topics:
  - C/C++ programming: Files and pointers.
  - Automata, Regular Expressions and Context-Free Grammar.
  - Data Structures: Trees, Hash tables, Linked lists, Stacks and Queues.
  - Knowledge of Linux commands is recommended.

You will need to use your laptop running a Linux distribution (Ubuntu 24.04) or Windows with WSL (Windows Subsystem for Linux)

10